

CAUVERY HYDROLOGICAL DATA ANALYTICS: WATER LEVEL, FLOW & RESERVOIR PERFORMANCE

1. Project Overview and Objective:

- **Project Overview:** This project involves cleaning, transforming, and analysing multi-source hydrological and agricultural datasets of the Cauvery River basin using Python (Google Colab) and creating interactive Power BI dashboards to derive meaningful insights on water availability and irrigation stress.
- **Objective:**
 - To analyse reservoir storage, river discharge, and water levels across Karnataka and Tamil Nadu.
 - To quantify agricultural water deficit in the Cauvery delta districts.
 - To compare upstream (Karnataka) vs downstream (Tamil Nadu) water availability.
 - To evaluate the growing dependence on groundwater due to insufficient canal supply.
 - To support data-driven reservoir release planning.

2. Data Sources:

- **Source:** India Water Resources Information System (India-WRIS), Central Water Commission (CWC), National Water Data Portal (NWDP), and Tamil Nadu agricultural data portals.
- **Timeline:** 2021 to 2026
- **Domain:** Hydrology / Water Resource Management & Agriculture Analytics.

3. Problem Statement:

- The Cauvery River basin faces seasonal water stress, causing irrigation shortages during critical crop seasons (Kuruva, Samba, and Thaladi) in Tamil Nadu's delta districts.
- This project analyses reservoir storage, river discharge, water levels, and agricultural demand to quantify water deficits, assess groundwater dependence, and compare upstream (Karnataka) versus downstream (Tamil Nadu) water availability for better water management.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Station / Reservoir Name	Text	Name of the monitoring station or reservoir
State	Categorical	Karnataka / Tamil Nadu / Puducherry
District	Categorical	Administrative district
Year	Numeric	Year of observation (2021–2026)
Season	Categorical	Winter / Southwest Monsoon / Northeast Monsoon etc.
Water_Level_Meters	Numeric	Recorded water level in meters
Daily Discharge (m3/s)	Numeric	Daily river discharge
Storage_Percentage	Numeric	Reservoir storage as percentage of capacity

Attribute Name	Data Type	Description
Inflow / Outflow	Numeric	Water inflow and outflow from reservoirs
Water_Demand_MCFT	Numeric	Agricultural water demand
Water_Supplied_MCFT	Numeric	Actual water supplied
Water_Deficit_MCFT	Numeric	Volume of water deficit
Deficit_Percentage	Numeric	Percentage of water deficit
Canal Supply Dependency %	Numeric	Percentage of water from canal
Groundwater Reliance %	Numeric	Percentage of water from groundwater

5. Tools & Technologies:

Python (Google Colab): Data cleaning, transformation, exploratory data analysis, and pre-processing.

Power BI: Data modelling, DAX measures, interactive dashboard creation, and visualization.

Excel: Initial data inspection and basic transformation.

6. Data Pre-Processing (Excel / Power Query):

➤ Tasks Performed:

- Removed null values and handled missing data in key columns (Date, Water Level, Discharge, etc.).
- Treated outliers in Inflow, Outflow, and Discharge using capping method.
- Created new columns such as Season, Basin Zone (Upstream/Downstream), Net Flow, and log transformations.
- Standardized date formats and extracted Year, Month, and Season.
- Merged and cleaned four datasets: Water Level, River Discharge, Reservoirs, and Agricultural Water Demand.
- Created calculated columns for better analysis (e.g., Storage Status, Discharge Category).

7. Data Modelling and DAX (Power BI):

Data Model:

- Created Dimension tables: Dim State and Dim Year.
- Established **Many-to-One** relationships between fact tables (Water Level, Discharge, Reservoirs, Agriculture) and Dimension tables.
- Used State and Year as common keys.

SCREENSHORTS OF DAX MEASURES:

Average Storage %

1 Avg Storage % = AVERAGE(cleaned_reservoirs[Storage_Percentage])

Date	Reservoir_Name	State	Structure_Type	Full_Depth_Feet	Capacity_MCFT	Water_Level_Feet	Storage_MCFT	Storage_Percentage
06 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
13 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
20 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
27 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
04 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
11 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
18 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
25 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
01 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
08 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
15 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
22 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
29 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
05 September 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
12 September 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
19 September 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
26 September 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
05 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
12 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
19 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
26 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
03 July 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10

Data

Search

- Cauvery River Flow
- cleaned_agriculture
- cleaned_discharge
- cleaned_reservoirs
 - Capacity_MCFT
- Date
 - Full_Depth_Feet
 - Inflow
 - Inflow (log)
- Measure
 - Month
 - Net_Flow (log)
 - Net_Flow_Cusecs
 - Outflow
 - Outflow (log)
 - Reservoir_Name
 - Season
 - State

Average Discharge %

1 Avg Discharge = AVERAGE(cleaned_discharge[Daily Discharge (m3/s)])

SlNo	Station	Agency	State LGD Code	State	District LGD Code	District	Tehsil	River	Basin	Tributary	Subtributary	Local River
264588	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264589	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264590	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264591	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264595	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264596	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264597	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264607	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264612	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264613	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264616	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264617	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264618	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264619	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264620	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264621	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264622	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264623	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264624	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264625	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264626	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi
264627	THIMMANAHALLI CWC		29	Karnataka	539	Hassan	HASSAN	Cauvery	Cauvery	Yagachi	-	Yagachi

Data

Search

- Cauvery River Flow
- cleaned_agriculture
- cleaned_discharge
 - Agency
 - Basin
 - Daily Discharge (log)
 - Daily Discharge (m3/s)
 - Discharge Category
 - Discharge Change (%)
 - Discharge Change (log)
 - Discharge Change (m3/s)
 - District
 - District LGD Code
 - Latitude
 - Local River
 - Longitude
- Measure
 - Month
 - Month Number

Average Water Level

1 Avg Water Level = AVERAGE(cleaned_water_level[Water_Level_Meters])

Station	State	District LGD Code	District	Tehsil	Basin	Tributary	Local River	Latitude	Longitude	Zero Gauge Reference Level (m)
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	
SRI RANGAM	Tamil Nadu	591	Tiruchirappalli	TIRUCHIRAPPALLI	Cauvery	-	Cauvery	10.84722222	78.68888889	

Data

Search

- Cauvery River Flow
- cleaned_agriculture
- cleaned_discharge
- cleaned_reservoirs
- cleaned_water_level
 - Actual Gauge Reading (m)
 - Basin
 - District
 - District LGD Code
 - Latitude
 - Local River
 - Longitude
 - Mean Sea Level Reference (m)
- Measure
 - Month
 - Observation Time
 - Season
 - State

- KA Retention %

1 KA Retention % =

2 `DIVIDE(`

3 `CALCULATE(`

4 `COUNTROWS(cleaned_reservoirs),`

5 `cleaned_reservoirs[State] = "Karnataka",`

6 `cleaned_reservoirs[Net_Flow_Cusecs] > 0`

7 `),`

8 `CALCULATE(`

9 `COUNTROWS(cleaned_reservoirs),`

10 `cleaned_reservoirs[State] = "Karnataka"`

11 `)`

12 `)`

Data

Search

- > Cauvery River Flow
- > cleaned_agriculture
- > cleaned_discharge
- > cleaned_reservoirs
 - Avg Storage %
 - Capacity_MCF
- > Date
 - Full_Depth_Feet
 - Inflow
 - Inflow (log)
 - Measure
 - Month
 - Net_Flow (log)
 - Net_Flow_Cusecs
 - Outflow
 - Outflow (log)
 - Reservoir_Name
 - Season

Date	Reservoir_Name	State	Structure_Type	Full_Depth_Feet	Capacity_MCF	Water_Level_Feet	Storage_MCF	Storage_Percentage
06 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
13 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
20 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
27 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
04 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
11 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
18 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
25 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
01 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
08 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
15 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
22 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10

- TN Retention %

```

1 TN Retention % =
2 DIVIDE(
3     CALCULATE(
4         COUNTROWS(cleaned_reservoirs),
5         cleaned_reservoirs[State] = "Tamil Nadu",
6         cleaned_reservoirs[Net_Flow_Cusecs] > 0
7     ),
8     CALCULATE(
9         COUNTROWS(cleaned_reservoirs),
10        cleaned_reservoirs[State] = "Tamil Nadu"
11    )
12 )

```

✕
✓

🔍

➤

Cauvery River Flow
cleaned_agriculture
cleaned_discharge
cleaned_reservoirs

📊

Avg Storage %
Capacity_MCFCT

➤

Date
Full_Depth_Feet
Inflow
Inflow (log)
KA Retention %
Measure
Month
Net_Flow (log)
Net_Flow_Cusecs
Outflow
Outflow (log)

Date	Reservoir_Name	State	Structure_Type	Full_Depth_Feet	Capacity_MCFCT	Water_Level_Feet	Storage_MCFCT	Storage_Percentage
06 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
13 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
20 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
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08 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
15 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
22 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10

- Upstream Storage %

X	✓
1	Upstream Storage % =
2	CALCULATE()
3	AVERAGE(cleaned_reservoirs[Storage_Percentage]),
4	cleaned_reservoirs[State] = "Karnataka"
5	

Date	Reservoir_Name	State	Structure_Type	Full_Depth_Feet	Capacity_MCFt	Water_Level_Feet	Storage_MCFt	Storage_Percentage
06 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
13 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
20 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
27 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
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19 September 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
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05 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
12 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10

- Downstream Storage %

1 Downstream Storage % =
2 CALCULATE(
3 AVERAGE(cleaned_reservoirs[Storage_Percentage]),
4 cleaned_reservoirs[State] = "Tamil Nadu"
5)

Date	Reservoir_Name	State	Structure_Type	Full_Depth_Feet	Capacity_MCFT	Water_Level_Feet	Storage_MCFT	Storage_Percentage
06 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
13 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
20 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
27 June 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
04 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
11 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
18 July 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
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08 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
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22 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
29 August 2021	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
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05 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10
12 June 2022	KALLANAI	Tamil Nadu	Barrage / Diversion Structure	0	0	0	0	10

- Total Water Deficit

1 Total Deficit = SUM(cleaned_agriculture[Water_Deficit_MCFT])

Build visuals with your data
Select or drag fields from the Data pane onto the report canvas.

- Average Deficit %

1 Avg Deficit % = AVERAGE(cleaned_agriculture[Deficit_Percentage])

Year	District	State	River_Basin_Zone	Season	Months	Major_Crop	Cultivated_Area_Hectares	Water_Demand_MCFT	Water_Supplied_MCFT	Water_Deficit_MCFT
2021	Cuddalore	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	19777	31496.1	29865.72	1630.38
2021	Cuddalore	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	43370	86399.12	81401.56	4997.56
2021	Cuddalore	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	7341	4044.08	3780.6	263.48
2021	Cuddalore	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	9641	14637.37	13645.09	1092.28
2021	Mayiladuthurai	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	24512	38526.02	36413.44	2112.58
2021	Mayiladuthurai	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	54671	107609.24	101276.17	6333.07
2021	Mayiladuthurai	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	8892	4734.71	4458.56	276.15
2021	Mayiladuthurai	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	11918	18234.84	17070.07	1164.77
2021	Nagapattinam	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	21903	36232.41	33872.52	2359.89
2021	Nagapattinam	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	48469	98821.61	92063.41	6758.2
2021	Nagapattinam	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	8046	4494.7	4260.09	234.61
2021	Nagapattinam	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	11118	16107.29	15012.11	1095.18
2021	Thanjavur	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	39498	66123.04	62564.29	3558.75
2021	Thanjavur	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	88868	175372.36	163787.24	11585.12
2021	Thanjavur	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	14291	7443.09	7080.13	362.96
2021	Thanjavur	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	19116	29157.24	27506.9	1650.34
2021	Thiruvannamalai	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	36164	57337.32	53581.08	3756.24
2021	Thiruvannamalai	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	74562	148798.64	140135	8663.64
2021	Thiruvannamalai	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	11008	6378.15	5901.48	476.67

- Average Canal %::

1 Avg Canal % = AVERAGE(cleaned_agriculture(Canal Supply Dependency X))										Data
Year	District	State	River_Basin_Zone	Season	Months	Major_Crop	Cultivated_Area_Hectares	Water_Demand_MCFY	Water_Supplied_MCFY	Water_Deficit_Percentage
2021	Cuddalore	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	19777	31496.1	29865.72	
2021	Cuddalore	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	43370	86399.12	81401.56	
2021	Cuddalore	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	7341	4044.08	3780.6	
2021	Cuddalore	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	9641	14637.37	13645.09	
2021	Mayiladuthurai	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	24512	38526.02	36413.44	
2021	Mayiladuthurai	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	54671	107609.24	101276.17	
2021	Mayiladuthurai	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	8892	4734.71	4458.56	
2021	Mayiladuthurai	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	11918	18234.84	17070.07	
2021	Nagapattinam	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	21903	36232.41	33872.52	
2021	Nagapattinam	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	48469	98821.61	92063.41	
2021	Nagapattinam	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	8046	4494.7	4260.09	
2021	Nagapattinam	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	11118	16107.29	15012.11	
2021	Thanjavur	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	39498	66123.04	62564.29	
2021	Thanjavur	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	88868	175372.36	163787.24	
2021	Thanjavur	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	14291	7443.09	7080.13	
2021	Thanjavur	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	19116	29157.24	27506.9	
2021	Thiruvannur	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	36164	57337.32	53581.08	
2021	Thiruvannur	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	74562	148798.64	140135	
2021	Thiruvannur	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	11808	6278.15	5891.48	

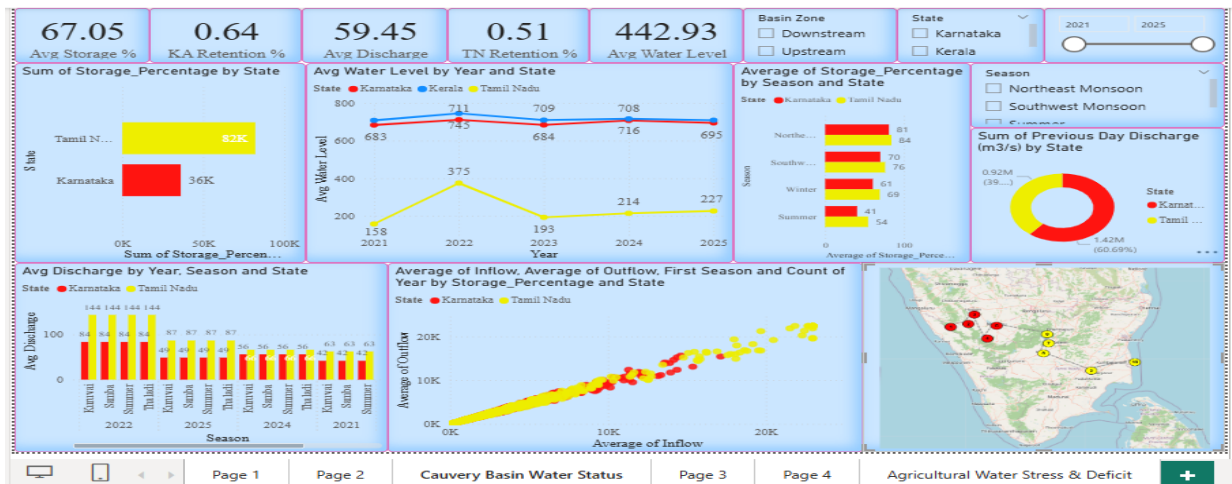
- Average Groundwater %

Avg GM % = AVERAGE(cleaned_agriculture[Groundwater Reliance %])										Data
Year	District	State	River_Basin_Zone	Season	Months	Major_Crop	Cultivated_Area_Hectares	Water_Demand_McFT	Water_Supplied_McFT	Water_Deficit_McFT
2021	Cuddalore	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	19777	31496.1	29865.72	
2021	Cuddalore	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	43370	86399.12	81401.56	
2021	Cuddalore	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	7341	4044.08	3780.6	
2021	Cuddalore	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	9641	14637.37	13645.09	
2021	Mayiladuthurai	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	24512	38526.02	36413.44	
2021	Mayiladuthurai	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	54671	107609.24	101276.17	
2021	Mayiladuthurai	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	8892	4734.71	4458.56	
2021	Mayiladuthurai	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	11918	18234.84	17070.07	
2021	Nagapattinam	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	21903	36232.41	33872.52	
2021	Nagapattinam	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	48469	98821.61	92063.41	
2021	Nagapattinam	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	8046	4494.7	4260.09	
2021	Nagapattinam	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	11118	16107.29	15012.11	
2021	Tanjavur	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	39498	66123.04	62564.29	
2021	Tanjavur	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	88868	175372.36	163787.24	
2021	Tanjavur	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	14291	7443.09	7080.13	
2021	Tanjavur	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	19116	29157.24	27506.9	
2021	Thiruvavur	Tamil Nadu	Delta	Kuruvai	Jun-Sep	Paddy (Rice)	36164	57337.32	53581.08	
2021	Thiruvavur	Tamil Nadu	Delta	Samba	Oct-Jan	Paddy (Rice)	74562	148798.64	140135	
2021	Thiruvavur	Tamil Nadu	Delta	Summer	Feb-May	Pulses/Gingelly	11808	6278.15	5891.48	
2021	Thiruvavur	Tamil Nadu	Delta	Thaladi	Oct-Jan	Paddy (Rice)	17381	25411.63	23979.1	

8. Analysis and Visualizations (Power BI)

Dashboard 1 – Cauvery Basin Water Status (Supply Side)

- KPI : Average Storage %, Average Discharge, Average Water Level, Retention %.
- Line Chart: Water Level Trend by Year
- Column Chart: Year-wise Average Discharge
- Scatter Chart: Inflow vs Outflow
- Map: Cauvery River Flow (Talacauvery to Poompuhar)
- Slicers: Year, State, Season, Basin Zone



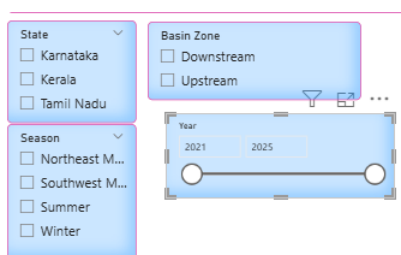
Dashboard 2 – Agricultural Water Stress & Deficit (Demand Side)

- KPI Cards: Total Deficit, Average Deficit %, Canal %, Groundwater %
- Donut Chart: Deficit Contribution by District
- Tree Map: Deficit by District and Season
- Stacked Area Chart: Demand vs Supplied vs Groundwater
- Gauge Chart: Average Deficit %
- Slicers: Year, District, Season, Crop

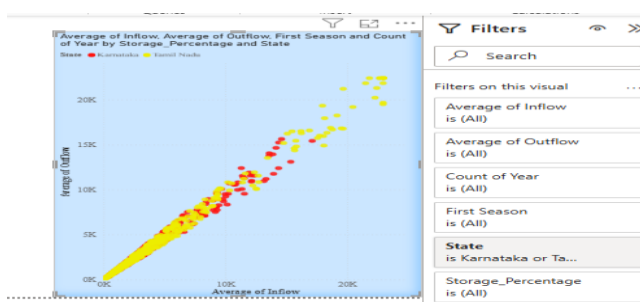


Interactive Features:

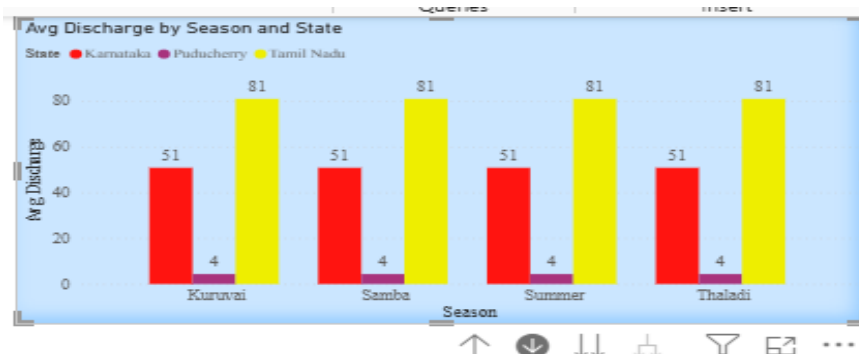
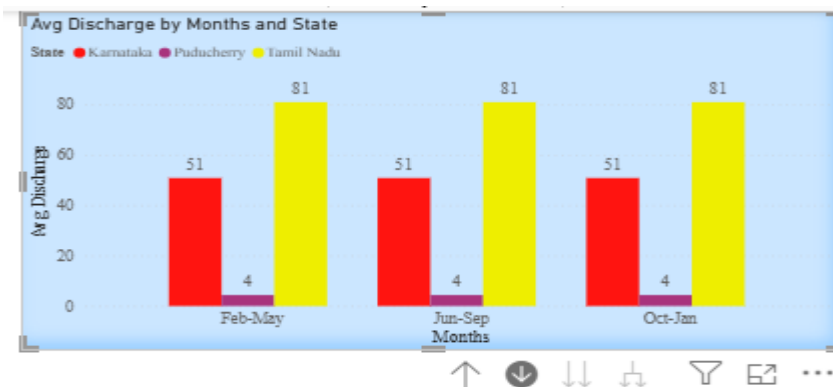
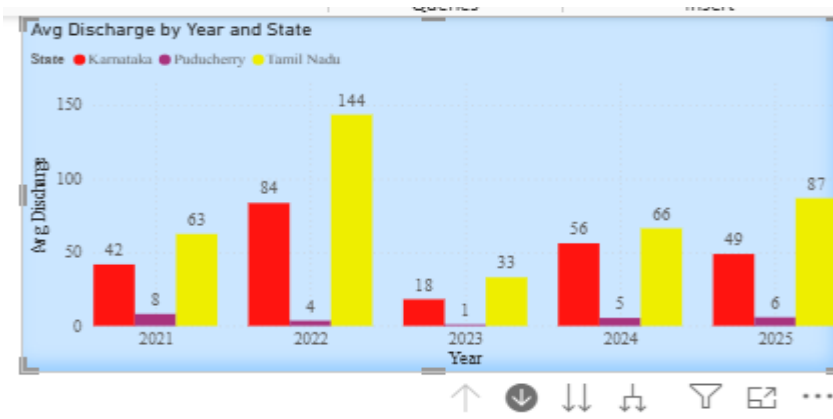
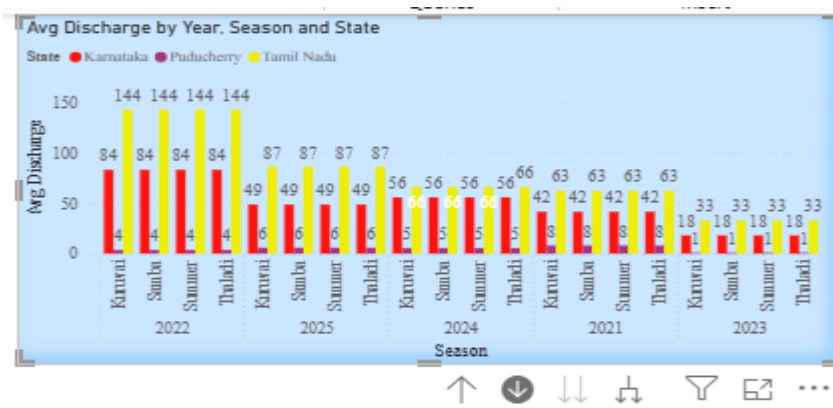
- Slicers



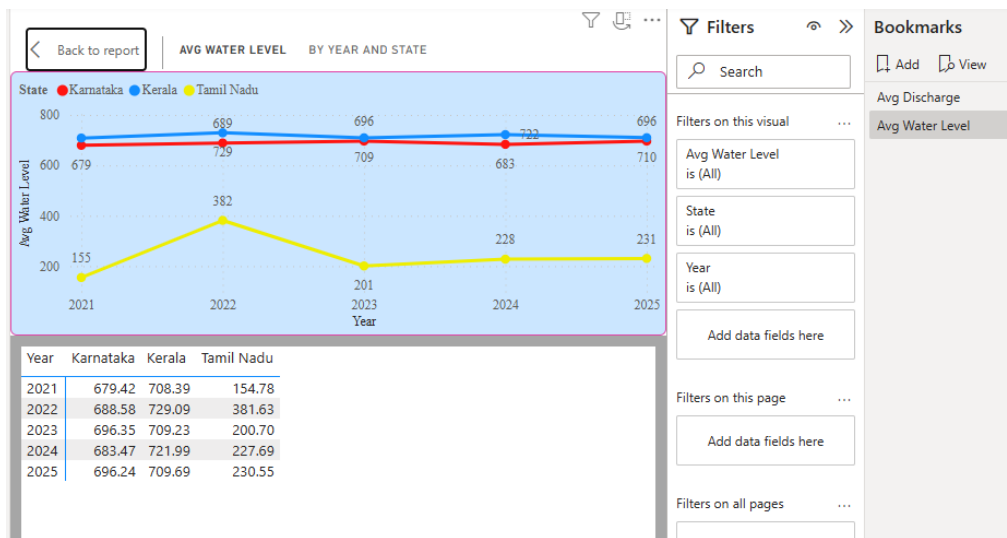
- Filters



- Drill-down



- Bookmarks.



9. Insights & Conclusions

Key Findings:

- Karnataka (Upstream) generally maintains higher reservoir storage than Tamil Nadu (Downstream).
- Significant water deficit is observed in delta districts, especially during Samba and Kuruvai seasons.
- Groundwater reliance has increased over the years to compensate for insufficient canal supply.
- Thanjavur, Thiruvarur, and Nagapattinam are among the most affected districts.

Analysis Insights:

- **Descriptive:** Overall storage, discharge, and deficit patterns across years and states.
- **Diagnostic:** Upstream water retention contributes to downstream irrigation stress.
- **Predictive:** Continued groundwater dependence may lead to long-term sustainability issues.
- **Prescriptive:** Better timed reservoir releases from Karnataka can reduce delta water stress.

10. Conclusion

The analysis of Cauvery basin hydrological and agricultural data reveals a clear upstream-downstream imbalance. Karnataka generally maintains higher reservoir storage, while Tamil Nadu's delta districts face recurring irrigation deficits, especially during critical cropping seasons. Increasing dependence on groundwater highlights the need for better-timed and coordinated reservoir releases to improve agricultural water security in the downstream region.

